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Clinical decision support of radiotherapy treatment planning: A data-driven machine learning strategy for patient-specific dosimetric decision making.

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CLINICAL & AGENTIC AI TAKEAWAY

This study developed a machine learning-based clinical decision support system that helps clinicians identify achievable historical treatment plans for new patients, enabling early dose tradeoffs and optimizing treatment for individual patients.

ABSTRACT

Keywords: Decision support system, Knowledge-based Planning, Dosimetric Tradeoffs

Study Overview & Background

Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

This study developed a machine learning-based clinical decision support system that helps clinicians identify achievable historical treatment plans for new patients, enabling early dose tradeoffs and optimizing treatment for individual patients.

Extended Repository Abstract

Background and purpose Clinical decision support systems are a growing class of tools with the potential to impact healthcare. This study investigates the construction of a decision support system through which clinicians can efficiently identify which previously approved historical treatment plans are achievable for a new patient to aid in selection of therapy. Material and methods Treatment data were collected for early-stage lung and postoperative oropharyngeal cancers

treated using photon (lung and head and neck) and proton (head and neck) radiotherapy. Machine-learning classifiers were constructed using patient-specific feature-sets and a library of historical plans. Model accuracy was analyzed using learning curves, and historical treatment plan matching was investigated. Results Learning curves demonstrate that for these datasets, approximately 45, 60, and 30 patients are needed for a sufficiently accurate classification model for radiotherapy for early-stage lung, postoperative oropharyngeal photon, and postoperative oropharyngeal proton, respectively. The resulting classification model provides a database of previously approved treatment plans that are achievable for a new patient. An exemplary case, highlighting tradeoffs between the heart and chest wall dose while holding target dose constant in two historical plans is provided. Conclusions We report on the first artificial-intelligence based clinical decision support system that connects patients to past discrete treatment plans in radiation oncology and demonstrate for the first time how this tool can enable clinicians to use past decisions to help inform current assessments. Clinicians can be informed of dose tradeoffs between critical structures early in the treatment process, enabling more time spent on finding the optimal course of treatment for individual patients.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

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